

Patrick Semler

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EDUCATION

Texas A&M University

Bachelor of Science in Computer Science

College Station, TX

Expected May 2028

- **GPA:** 3.79/4.00 | **Honors:** National Merit Scholar; President's Endowed Scholar
- **Coursework:** Program Design and Concepts; Data Structures and Algorithms; Computer Organization; Programming Languages; Discrete Math; Linear Algebra

EXPERIENCE

Research Software Engineer

Texas A&M University

Aug. 2026 – Present

College Station, TX

- Built an interactive web platform that replaced a spreadsheet prototype for simulating and visualizing U.S. beef supply-chain scenarios
- Modeled independent businesses across cow-calf, stocker, feedlot, and packer/retail stages, incorporating mortality, biological variation, inventory, market prices, and sensitivity analysis

Machine Learning Research Engineer

Texas A&M University

June 2026 – Aug. 2026

College Station, TX

- Reduced a random-forest model's peak memory usage from 2.6 GB to 689 MB and its size by 4×, enabling Vercel deployment while preserving 90.83% accuracy versus 91.06%
- Built a Python/Django RAG pipeline using Voyage embeddings, Supabase pgvector, and ranked evidence retrieval to ground Gemini veterinary care plans in cited sources
- Consolidated separate disease-prediction, Gemini assistant, and SOAP-note interface iterations into one veterinary diagnostic site for AI-assisted assessments and treatment plans

Full-Stack Developer

ApplyFinch

Jan. 2026 – Present

College Station, TX

- Built a multi-signal job-matching and application-generation pipeline for a platform with 200+ registered users, evaluating opportunities using job and profile data, explaining match reasoning, and generating tailored resumes and cover letters on demand
- Made asynchronous application generation recoverable by saving run snapshots in Supabase, retrying failed database writes, preventing duplicate applications, and coordinating cancellation across workers
- Rebuilt React/TypeScript onboarding and profile editing with a confirmation workflow so users could review and correct LinkedIn and resume data before it powered matching and document generation

PROJECTS

LeetMind | *TypeScript, Python, React 19, FastAPI, PostgreSQL, Docker*

2026

- Built a full-stack adaptive coding platform that selects and generates programming problems based on each learner's skill level and topic history
- Validated AI-generated problems with authored tests, an independent test implementation, and 50 reproducible randomized inputs, publishing only problems that passed every quality check
- Ran Python and C++ submissions in isolated, network-disabled Docker containers while keeping hidden tests and expected answers outside the user's execution environment

LiftLedger | *TypeScript, Node.js, Fastify, React 19, PostgreSQL, Docker*

2026

- Built a React dashboard and Telegram fitness assistant that converts natural-language requests into structured workout and nutrition logs, tracking exercises, sets, calories, and macronutrients through a Fastify/PostgreSQL backend
- Turned Telegram voice messages and food photos into structured calorie and macronutrient records through approval-gated task graphs, allowing users to review updates before adding them to their tracking history
- Integrated Hevy workout history into progress dashboards, graphing set counts and training trends over time across exercises and muscle groups

TECHNICAL SKILLS

Languages: TypeScript, Python, C++, SQL

Web and APIs: React, FastAPI, Flask, Django, Fastify, Node.js, REST APIs

Data and Infrastructure: PostgreSQL, Supabase, Docker, Cloudflare Workers, Vercel

Machine Learning: scikit-learn, ONNX Runtime

AI Engineering: LLM Application Development, RAG (Retrieval-Augmented Generation), AI Agent Development

Testing and Tools: Git, GitHub, pytest, Vitest, Playwright, Testing Library